Signals and Systems

Z-Transform

(lecture 28, Date:-13/04/2026)

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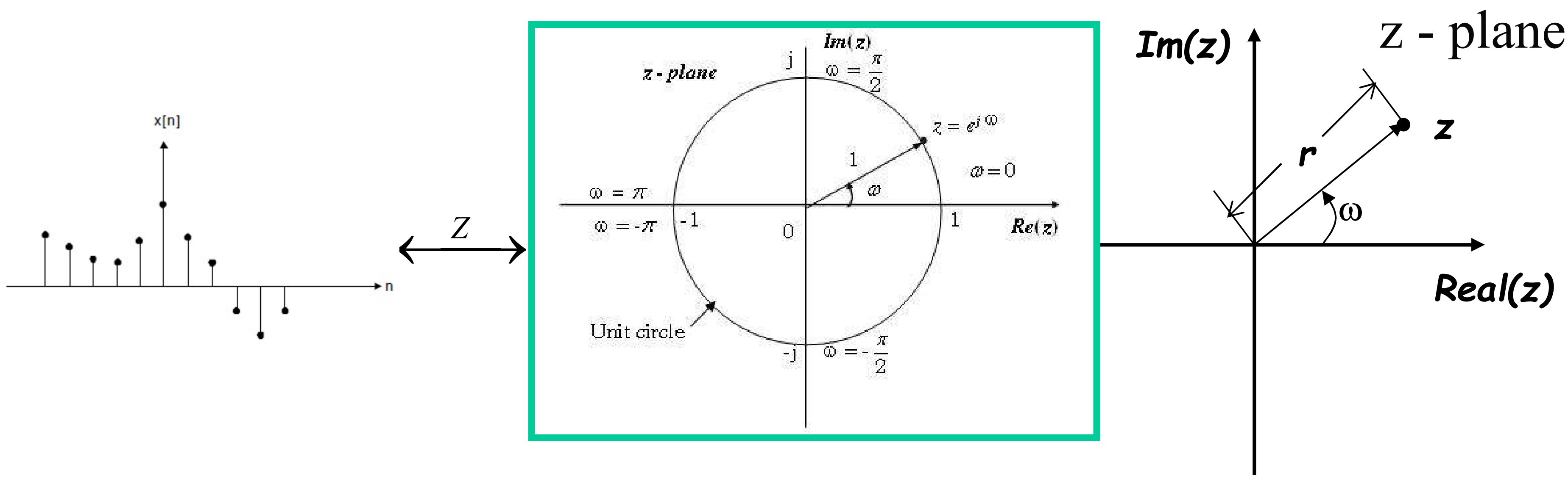
Z-Transform

❖In Z-Transform, the discrete time signal x(n) is transformed from time domain into complex Z-plane .

❖Z-Transform of a the discrete time signal x(n) is defined as power series

$$Z[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$ Where z is complex variable



z is a complex variable: $z = re^{j\omega}$

❖Z-Transform of a the discrete time signal x(n) is denoted by Z[x(n)] or

$$x(n) \xleftrightarrow{Z} X(z)$$

ROC (Region of Convergence)

❖ROC of X(z) is the set of all values of z for which X(z) (power series defined above) attains finite value

❖X(z) exists for those value of z for power series (X(z) is infinite power series) converges.

❖i.e. X(z) exists only for its ROC.

Example 1: Find the Z-transform of following signals .

(i) $x[n] = \{1, 2, 5, 7, 0, 1\}$

(ii) $x[n] = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$

(iii) $x[n] = \delta(n)$

(iv) $x[n] = \delta(n-k)$

(v) $x[n] = \delta(n+k)$

(vi) $x[n] = a^n u[n]$

(vii) $x[n] = -a^n u[-n-1]$

(viii) $x[n] = a^n u(n) + b^n u[-n-1]$

(ix) $x[n] = a^n u[-n]$

(x) $x[n] = 7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n]$

(xi) $x[n] = 7\left(\frac{1}{3}\right)^n u[-n-1] - 6\left(\frac{1}{2}\right)^n u[-n-1]$

(xii) $x[n] = 7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[-n-1]$

(xiii) $x[n] = 7\left(\frac{1}{3}\right)^n u[-n-1] - 6\left(\frac{1}{2}\right)^n u[n]$

(xiv) $x[n] = \sin(\omega_0 n) u[n]$

(xv) $x[n] = \cos(\omega_0 n) u[n]$

(xvi) $x[n] = b^{|n|}, , b > 0$

(i) $x[n]=\{1,2,5,7,0,1\}$

Sol:-

$$X(z)=\sum_{n=-\infty}^{\infty}x[n]z^{-n}$$

$$=+.....+x(-2)z^{2}+x(-1)z+x(0)+x(1)z^{-1}+x(2)z^{-2}+....$$

$$=1+2z^{-1}+5z^{-2}+7z^{-3}+z^{-5}$$

ROC of X(z): entire z-plane except z=0

(ii) $x[n]=\{1,2,\underset{\uparrow}{5},7,0,1\}$

Sol:-

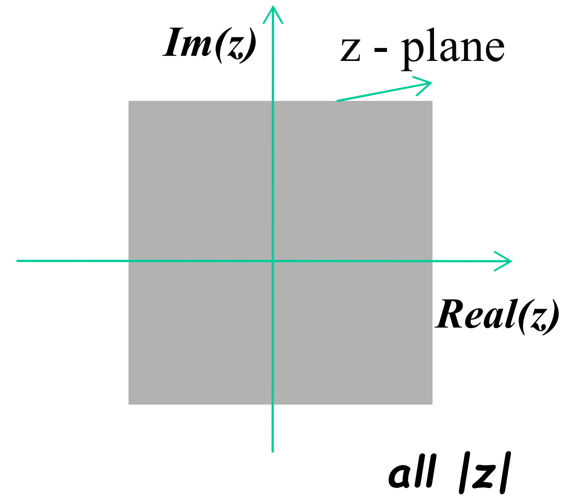
$$X(z)=\sum_{n=-\infty}^{\infty}x[n]z^{-n}=z^{2}+2z+5+7z^{-1}+z^{-3}$$

ROC of X(z): entire z-plane except z=0 and z = infinite

(iii) $x[n] = \delta(n)$

Sol:-

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \sum_{n=-\infty}^{\infty} \delta[n]z^{-n} = 1$$



ROC of X(z): entire z-plane

(iv) $x[n] = \delta(n-k)$

Sol:-

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \sum_{n=-\infty}^{\infty} \delta(n-k)z^{-n} = z^{-k}$$

ROC of X(z): entire z-plane except z=0

(v) $x[n] = \delta(n+k)$

Sol:-

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \sum_{n=-\infty}^{\infty} \delta(n+k)z^{-n} = z^{k}$$

ROC of X(z): entire z-plane except z= infinite

(vi) $x[n] = a^n u[n]$ $\longrightarrow$ Causal signal

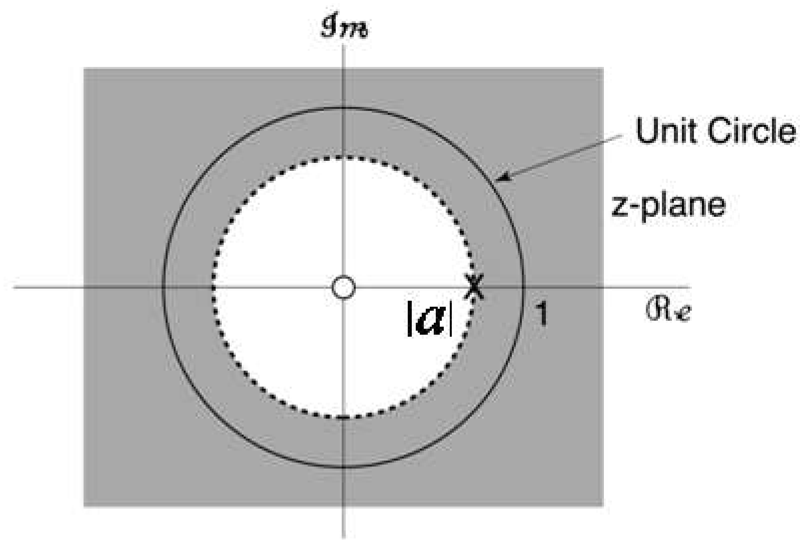
Sol:-

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= 1 + az^{-1} + (az^{-1})^2 + (az^{-1})^3 + (az^{-1})^4 + \ldots\ldots = \frac{1}{1 - az^{-1}}$$

ROC of X(z): $\left|az^{-1}\right| < 1 \Rightarrow |z| > |a|$

ROC of X(z) will be the exterior of circle having radius $|a|$ in z-plane.



(vii) $x[n] = -a^n u[-n-1]$ ⟶ Anti Causal signal

$$= -a^n ,,, n \le -1$$

$$= 0,,, n \ge 0$$

Sol:-

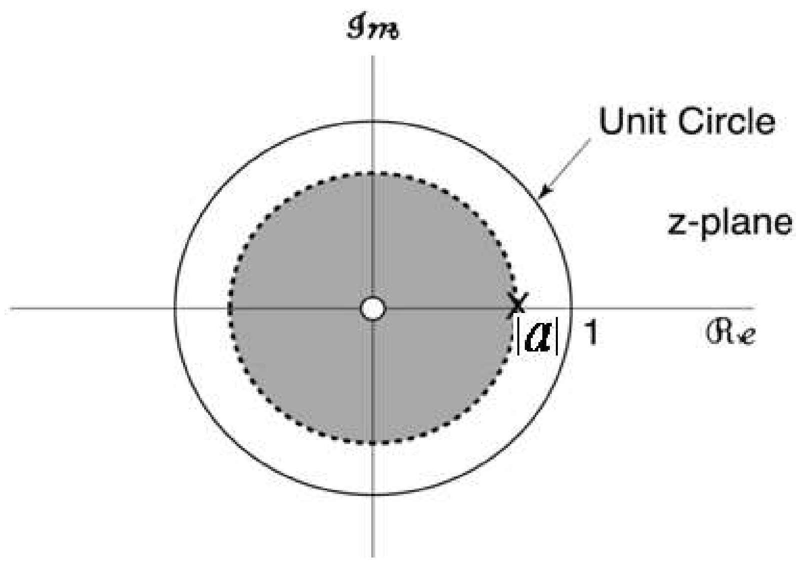
$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = \sum_{n=-\infty}^{-1} -a^n z^{-n} = \sum_{n=1}^{\infty} -a^{-n} z^n$$

$$= -a^{-1}z - (a^{-1}z)^2 - (a^{-1}z)^3 - (a^{-1}z)^4 - (a^{-1}z)^5 - \ldots\ldots$$

$$= -a^{-1}z[1 + (a^{-1}z) + (a^{-1}z)^2 + (a^{-1}z)^3 + (a^{-1}z)^4 + \ldots\ldots]$$

$$= \frac{-a^{-1}z}{1-a^{-1}z} = \frac{-1}{az^{-1}-1} = \frac{1}{1-az^{-1}}$$

ROC of X(z): $\left|a^{-1}z\right| < 1 \Rightarrow |z| < |a|$



➢ROC of X(z) will be the interior of circle having radius $|a|$ in z-plane.

(viii) $x[n] = a^n u(n) + b^n u[-n-1] \longrightarrow$ Non Causal signal

Sol:- $$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = \sum_{n=-\infty}^{\infty} [a^n u(n) + b^n u(-n-1)] z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n} + \sum_{n=-\infty}^{-1} b^n z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n} + \sum_{n=1}^{\infty} b^{-n} z^{n}$$

Both Series Converge

$|z| > |a|$ $\quad$ $|z| < |b|$

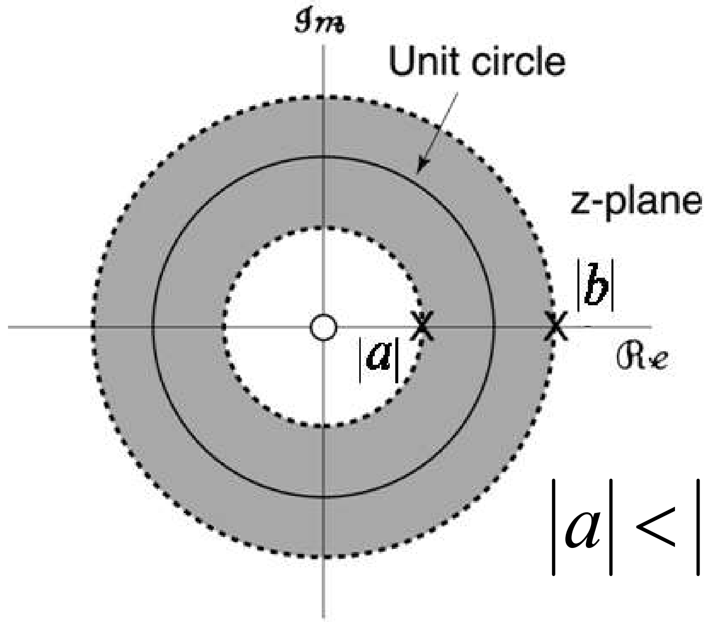
Above X(z) will converge if both series simultaneously converge

ROC of X(z): $(|z| > |a|) \cap (|z| < |b|)$

Case 1: when $|b| > |a|$ $\quad$ $$X(z) = \frac{1}{1 - az^{-1}} - \frac{1}{1 - bz^{-1}}$$

ROC of X(z): $|a| < |z| < |b|$

ROC of X(z) will be the intersection of two circles one having radius (smaller) $|a|$ other having radius (bigger) $|b|$



$$|a| < |z| < |b|$$

Case 2: when $|a| > |b|$ ROC of X(z): $|z| \in \phi$

X(z) will not exist .

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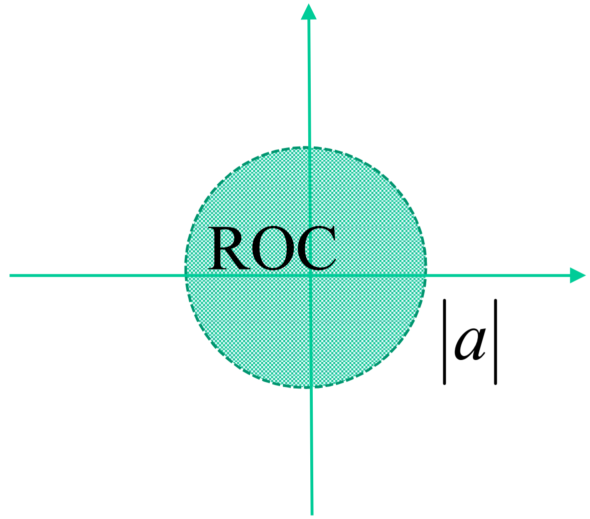
(ix) $x[n] = a^n u[-n]$

Sol:-

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = \sum_{n=-\infty}^{0} a^n z^{-n} = \sum_{n=0}^{\infty} \left(a^{-1} z\right)^n$$

$$= \frac{1}{1 - a^{-1} z}$$

ROC of X(z): $\left|a^{-1} z\right| < 1 \Rightarrow |z| < |a|$



(x) $x[n] = 7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n]$

Sol:-

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

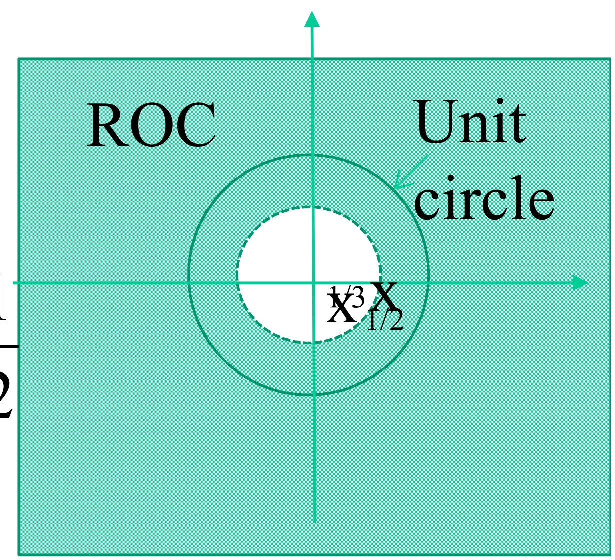
$$= \sum_{n=0}^{\infty} \left[7\left(\frac{1}{3} z^{-1}\right)^n - 6\left(\frac{1}{2} z^{-1}\right)^n \right]$$

$$= \frac{7}{1 - \frac{1}{3} z^{-1}} - \frac{6}{1 - \frac{1}{2} z^{-1}}$$

ROC of X(z):

$$|z| > \frac{1}{3} \cap |z| > \frac{1}{2}$$

$$\Rightarrow |z| > \frac{1}{2}$$



➢ROC of X(z) of any causal signal which is sum of two or more causal signal , will be the exterior of largest circle .

(xi) $$x[n] = 7\left(\frac{1}{3}\right)^n u[-n-1] - 6\left(\frac{1}{2}\right)^n u[-n-1]$$

Sol:-

$$7\left(\frac{1}{3}\right)^n u[-n-1] \xleftrightarrow{Z} -\frac{7}{1-\frac{1}{3}z^{-1}},,,,, ROC: |z| < \frac{1}{3}$$

$$-6\left(\frac{1}{2}\right)^n u[-n-1] \xleftrightarrow{Z} \frac{6}{1-\frac{1}{2}z^{-1}},,,,,, ROC: |z| < \frac{1}{2}$$

$$7\left(\frac{1}{3}\right)^n u[-n-1] - 6\left(\frac{1}{2}\right)^n u[-n-1] \xleftrightarrow{Z} -\frac{7}{1-\frac{1}{3}z^{-1}} + \frac{6}{1-\frac{1}{2}z^{-1}} \quad ROC: |z| < \frac{1}{3}$$

➢ROC of X(z) of any anti-causal signal which is sum of two or more anti-causal signal , will be the interior of smallest circle .

Examples:- $x[n]=7\left(\frac{1}{3}\right)^n u[n]-6\left(\frac{1}{2}\right)^n u[n]+3u[n]$

Sol:-

$$7\left(\frac{1}{3}\right)^n u[n] \xleftrightarrow{Z} \frac{7}{1-\frac{1}{3}z^{-1}},,,,,ROC:|z|>\frac{1}{3}$$

$$-6\left(\frac{1}{2}\right)^n u[n] \xleftrightarrow{Z} -\frac{6}{1-\frac{1}{2}z^{-1}},,,,,,ROC:|z|>\frac{1}{2}$$

$$3u[n] \xleftrightarrow{Z} \frac{3}{1-z^{-1}},,,,,,ROC:|z|>1$$

$$7\left(\frac{1}{3}\right)^n u[n]-6\left(\frac{1}{2}\right)^n u[n]+3u[n] \xleftrightarrow{Z} \frac{7}{1-\frac{1}{3}z^{-1}}-\frac{6}{1-\frac{1}{2}z^{-1}}+\frac{3}{1-z^{-1}} \quad ROC:|z|>1$$

Examples:- $$x[n]=7\left(\frac{1}{3}\right)^n u[-n-1]-6\left(\frac{1}{2}\right)^n u[-n-1]-3\left(\frac{1}{4}\right)^n u[-n-1]$$

Sol:-

$$7\left(\frac{1}{3}\right)^n u[-n-1]\xleftrightarrow{Z}-\frac{7}{1-\frac{1}{3}z^{-1}},,,,,ROC:|z|<\frac{1}{3}$$

$$-6\left(\frac{1}{2}\right)^n u[-n-1]\xleftrightarrow{Z}\frac{6}{1-\frac{1}{2}z^{-1}},,,,,,ROC:|z|<\frac{1}{2}$$

$$-3\left(\frac{1}{4}\right)^n u[-n-1]\xleftrightarrow{Z}\frac{3}{1-\frac{1}{4}z^{-1}},,,,,,ROC:|z|<\frac{1}{4}$$

$$7\left(\frac{1}{3}\right)^n u[-n-1]-6\left(\frac{1}{2}\right)^n u[-n-1]-3\left(\frac{1}{4}\right)^n u[-n-1]\xleftrightarrow{Z}\frac{7}{1-\frac{1}{3}z^{-1}}+\frac{6}{1-\frac{1}{2}z^{-1}}+\frac{3}{1-\frac{1}{4}z^{-1}}$$

$$ROC:|z|<\frac{1}{4}$$

(xii) $x[n]=7\left(\frac{1}{3}\right)^n u[n]-6\left(\frac{1}{2}\right)^n u[-n-1]$

Sol:-

$$7\left(\frac{1}{3}\right)^n u[n]\xleftrightarrow{Z}\frac{7}{1-\frac{1}{3}z^{-1}},,,,,ROC:|z|>\frac{1}{3}$$

$$-6\left(\frac{1}{2}\right)^n u[-n-1]\xleftrightarrow{Z}\frac{6}{1-\frac{1}{2}z^{-1}},,,,,,ROC:|z|<\frac{1}{2}$$

Since:-

$$(|z|>\frac{1}{3})\bigcap(|z|<\frac{1}{2})=\frac{1}{3}<|z|<\frac{1}{2}$$

$$7\left(\frac{1}{3}\right)^n u[n]-6\left(\frac{1}{2}\right)^n u[-n-1]\xleftrightarrow{Z}\frac{7}{1-\frac{1}{3}z^{-1}}+\frac{6}{1-\frac{1}{2}z^{-1}}\quad ROC:\frac{1}{3}<|z|<\frac{1}{2}$$

(xiii) $x[n] = 7\left(\frac{1}{3}\right)^n u[-n-1] - 6\left(\frac{1}{2}\right)^n u[n]$

Sol:-

$$7\left(\frac{1}{3}\right)^n u[-n-1] \xleftrightarrow{Z} -\frac{7}{1-\frac{1}{3}z^{-1}},,,,,ROC:|z|<\frac{1}{3}$$

$$-6\left(\frac{1}{2}\right)^n u[n] \xleftrightarrow{Z} -\frac{6}{1-\frac{1}{2}z^{-1}},,,,,,ROC:|z|>\frac{1}{2}$$

Since:- $$(|z|<\frac{1}{3}) \cap (|z|>\frac{1}{2}) = \phi(nullset)$$

So Z-transform of above signal X(z) does not exist..

(xiv) $x[n] = \sin(\omega_0 n)u[n]$

Sol:-

$$x[n] = \sin(\omega_0 n)u[n] = \frac{e^{j\omega_0 n} - e^{-j\omega_0 n}}{2j}u(n)$$

$$= \frac{1}{2j}(e^{j\omega_0})^n u(n) - \frac{1}{2j}(e^{-j\omega_0})^n u(n)$$

$$\frac{1}{2j}(e^{j\omega_0})^n u(n) \xleftrightarrow{Z} \frac{1}{2j}\frac{1}{1-e^{j\omega_0}z^{-1}} \qquad ROC: |z| > |e^{j\omega_0}| \Rightarrow |z| > 1$$

$$\frac{1}{2j}(e^{-j\omega_0})^n u(n) \xleftrightarrow{Z} \frac{1}{2j}\frac{1}{1-e^{-j\omega_0}z^{-1}} \qquad ROC: |z| > |e^{-j\omega_0}| \Rightarrow |z| > 1$$

$$X(z) = Z[\sin(\omega_0 n)u(n)] = \frac{1}{2j}\left[\frac{1}{1-e^{j\omega_0}z^{-1}} - \frac{1}{1-e^{-j\omega_0}z^{-1}}\right] \qquad ROC: |z| > 1$$

(xv) $x[n] = \cos(\omega_0 n)u[n]$

Sol:-

$$x[n] = \cos(\omega_0 n)u[n] \quad = \frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2}u(n)$$

$$= \frac{1}{2}(e^{j\omega_0})^n u(n) + \frac{1}{2}(e^{-j\omega_0})^n u(n)$$

$$\frac{1}{2}(e^{j\omega_0})^n u(n) \xleftrightarrow{Z} \frac{1}{2}\frac{1}{1-e^{j\omega_0}z^{-1}} \qquad ROC: |z| > |e^{j\omega_0}| \Rightarrow |z| > 1$$

$$\frac{1}{2}(e^{-j\omega_0})^n u(n) \xleftrightarrow{Z} \frac{1}{2}\frac{1}{1-e^{-j\omega_0}z^{-1}} \qquad ROC: |z| > |e^{-j\omega_0}| \Rightarrow |z| > 1$$

$$X(z) = Z[\cos(\omega_0 n)u(n)] \quad = \frac{1}{2}[\frac{1}{1-e^{j\omega_0}z^{-1}} + \frac{1}{1-e^{-j\omega_0}z^{-1}}] \quad ROC: |z| > 1$$

Examples:- $$x[n]=\left(\frac{1}{3}\right)^n \sin\left(\frac{\pi}{4}n\right)u[n]$$

Sol:-

$$x[n]=\left[\frac{1}{2j}\left(\frac{1}{3}\right)^n e^{j\left(\frac{\pi}{4}n\right)}-\frac{1}{2j}\left(\frac{1}{3}\right)^n e^{-j\left(\frac{\pi}{4}n\right)}\right]u[n]$$

$$x[n]=\left[\frac{1}{2j}\left(\frac{1}{3}e^{j\frac{\pi}{4}}\right)^n-\frac{1}{2j}\left(\frac{1}{3}e^{-j\frac{\pi}{4}}\right)^n\right]u[n]$$

$$X(z)=\frac{1}{2j}\left(\frac{1}{1-\frac{1}{3}e^{j\frac{\pi}{4}}z^{-1}}-\frac{1}{1-\frac{1}{3}e^{-j\frac{\pi}{4}}z^{-1}}\right) \qquad ROC: |z|>\left|\frac{1}{3}e^{\pm j\frac{\pi}{4}}\right| \Rightarrow |z|>\frac{1}{3}$$

$$\text{(xvi)} \quad x[n] = b^{|n|}, \quad b > 0$$

$$\text{Sol:-} \quad x[n] = b^n u[n] + b^{-n} u[-n-1]$$

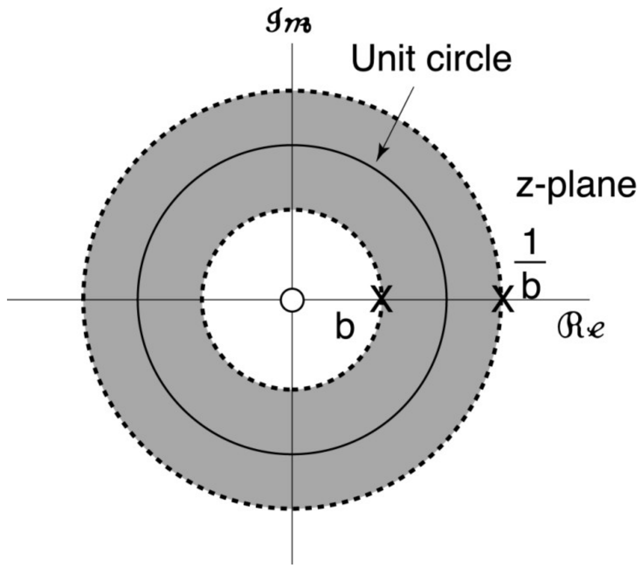
From:

$$b^n u[n] \longleftrightarrow \frac{1}{1 - bz^{-1}}, \quad |z| > b$$

$$b^{-n} u[-n-1] \longleftrightarrow \frac{-1}{1 - b^{-1}z^{-1}}, \quad |z| < \frac{1}{b}$$

$$X(z) = \frac{1}{1 - bz^{-1}} + \frac{-1}{1 - b^{-1}z^{-1}} \quad , \quad b < |z| < \frac{1}{b}$$

(if $b < 1$)



Clearly, ROC does *not* exist if $b > 1 \Rightarrow$ *No* z-transform for $b^{|n|}$.

Thank You